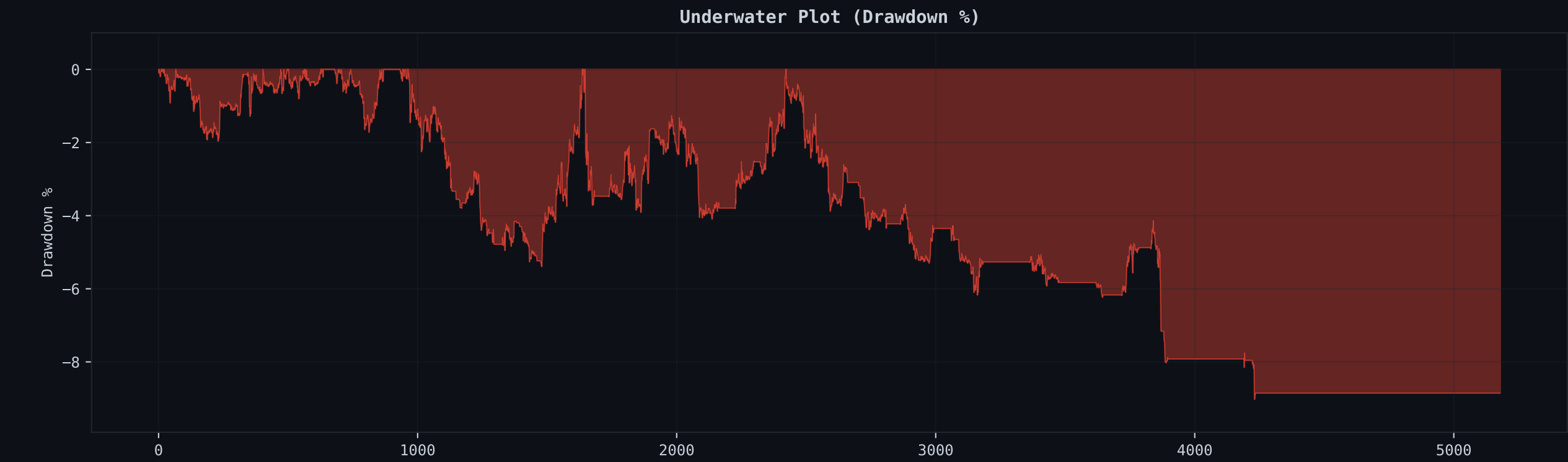
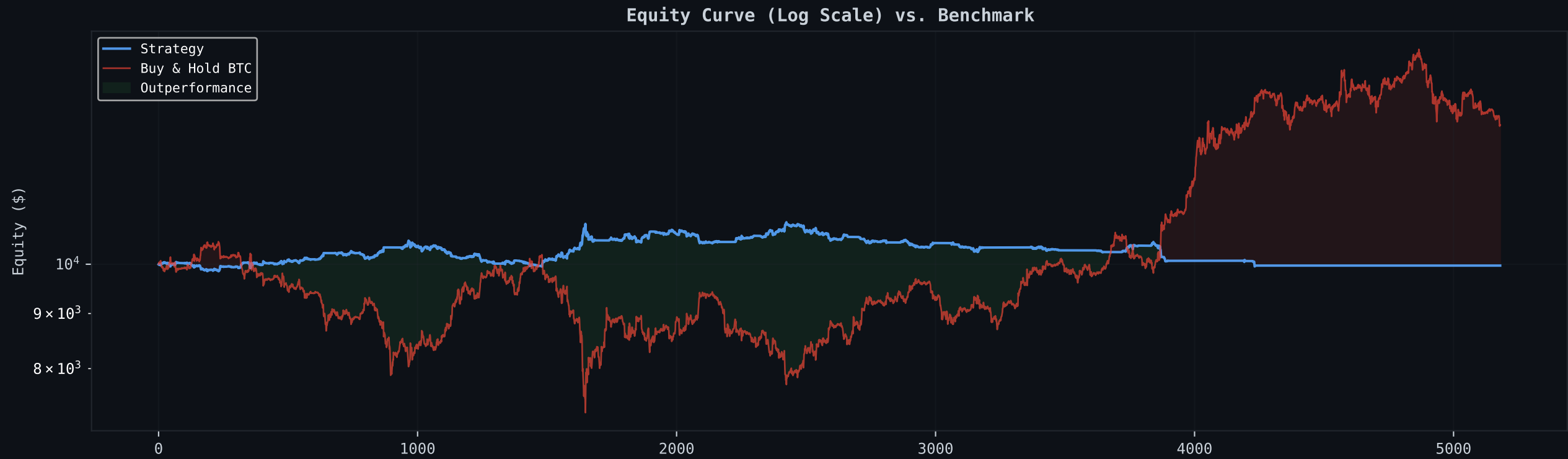


INSTITUTIONAL QUANTITATIVE TEARSHEET
CNN+LSTM PPO × SPA Day Trading Strategy

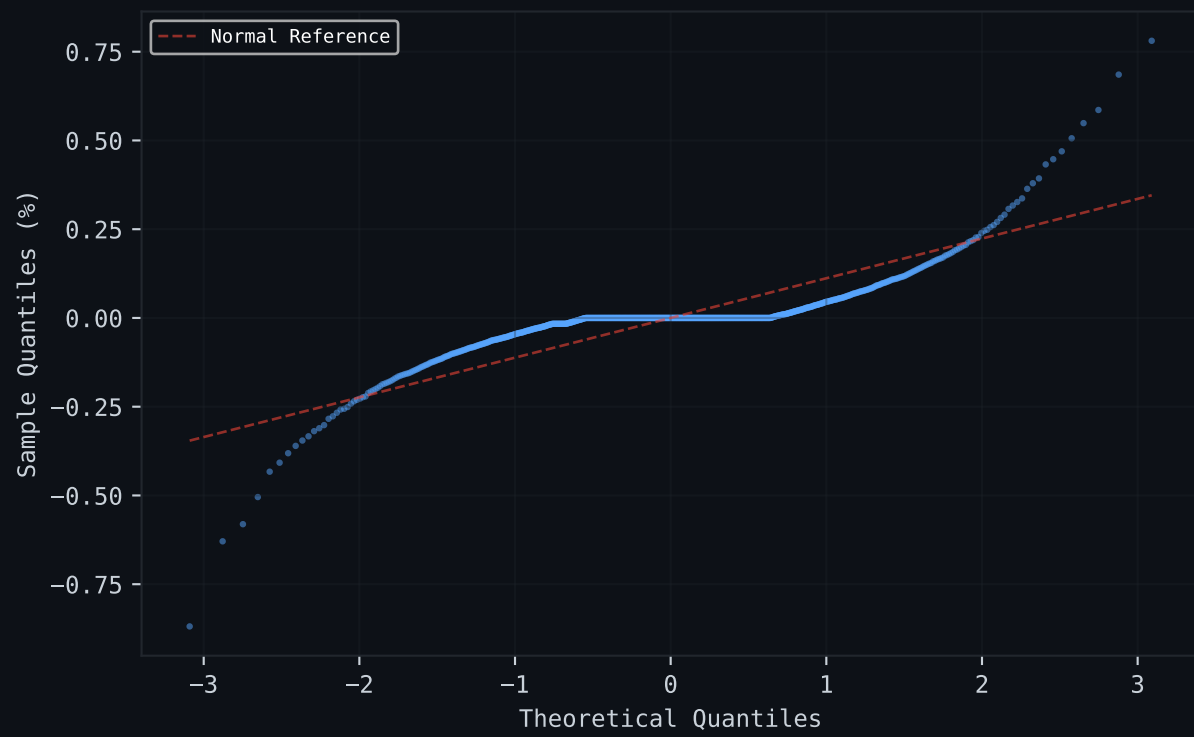
Sharpe: -0.04 | Sortino: -0.04 | Calmar: -0.05 | Max DD: -9.0% | Alpha p-value: 0.5184 | Beta: -0.145



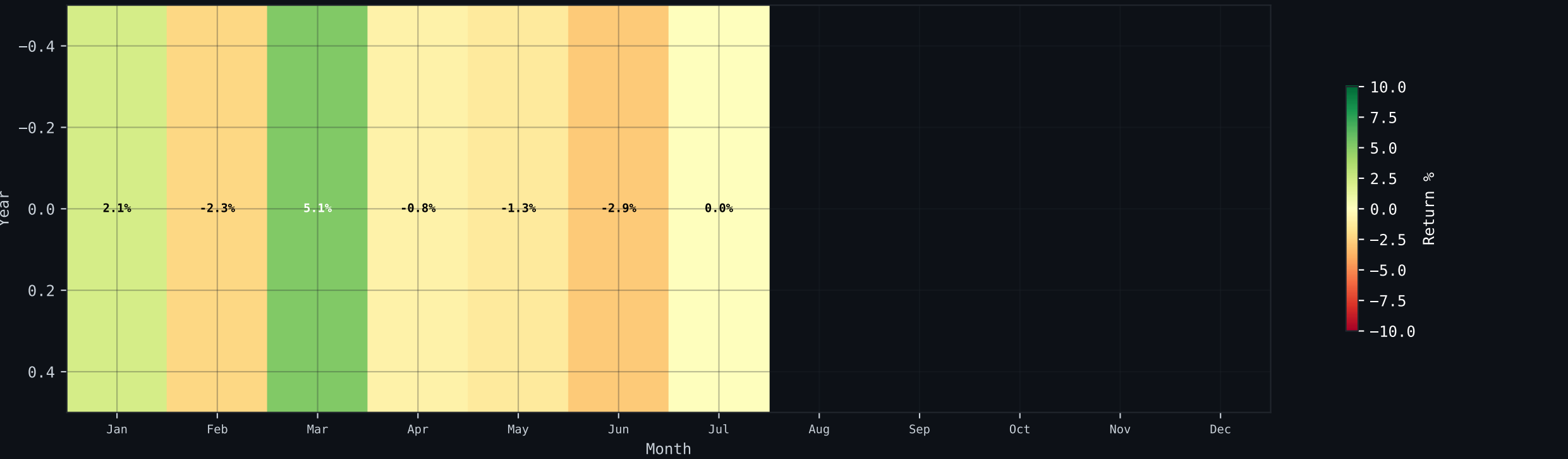
Return Distribution vs. Normal



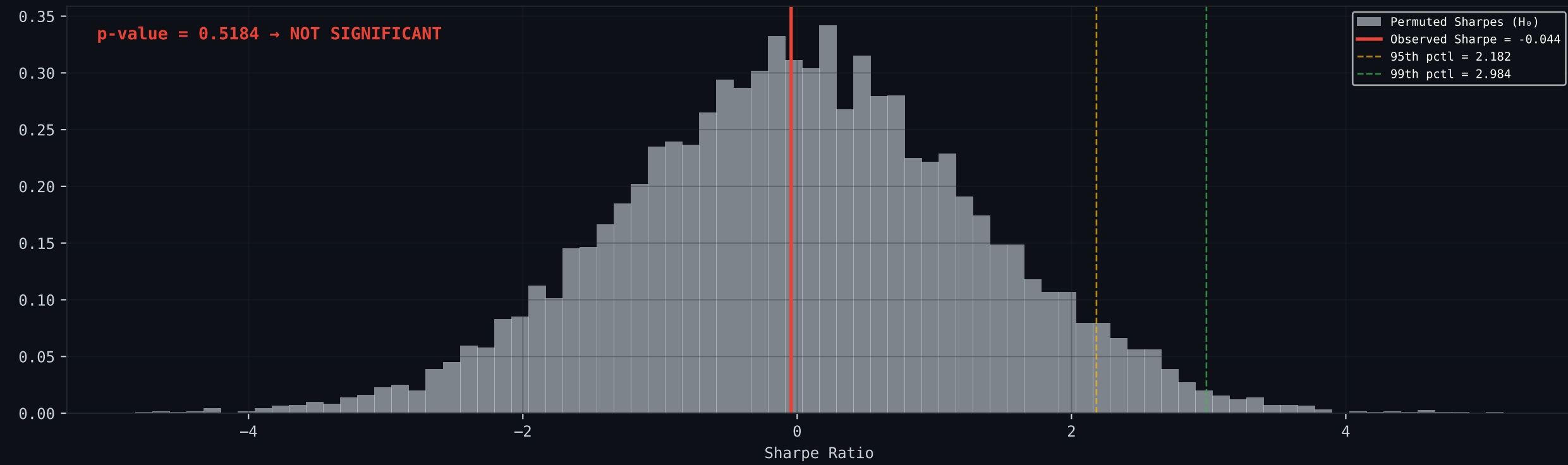
Q-Q Plot (Fat Tail Detection)



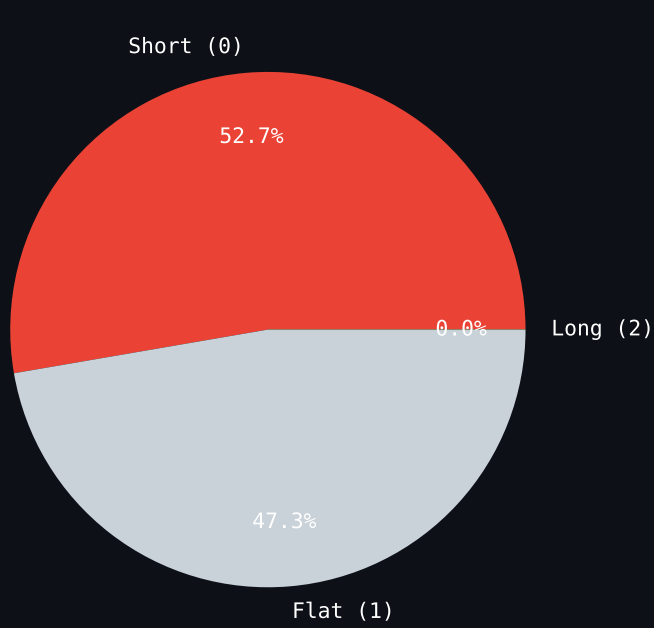
Monthly Returns Heatmap (%)



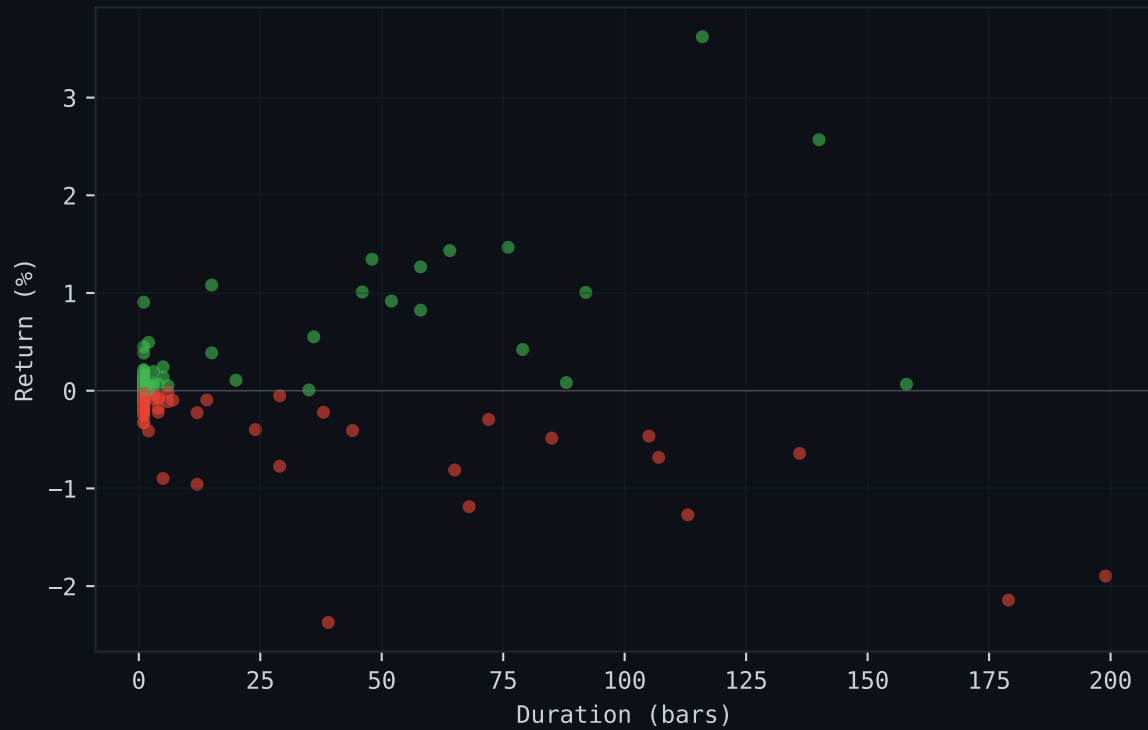
Monte Carlo Permutation Test (10,000 simulations)



Action Distribution



Trade Returns vs. Duration



QUANTITATIVE RISK METRICS – FULL DETAIL

Metric	Value
PERFORMANCE METRICS	
Total Return	-0.27%
Annualized Return	-0.46%
Annualized Volatility	10.47%
Sharpe Ratio	-0.044
Sortino Ratio	-0.043
Calmar Ratio	-0.051
Omega Ratio ($\theta=0$)	1.000
Profit Factor	1.000
Win Rate	47.1%
TAIL RISK METRICS	
Max Drawdown	-9.02%
VaR (95%)	-0.1474%
VaR (99%)	-0.3332%
CVaR / Expected Shortfall (95%)	-0.2745%
CVaR / Expected Shortfall (99%)	-0.5346%
Skewness	-0.159
Excess Kurtosis	21.638
ALPHA DECOMPOSITION	
Alpha (per period)	0.000011
Alpha t-statistic	0.949
Alpha p-value	0.3429
Beta (benchmark exposure)	-0.1447
R² (benchmark correlation)	0.4894
STATISTICAL VALIDATION	
Permutation Test p-value	0.5184
Significance ($p < 0.05$)	NO ☐
CAPACITY ESTIMATION	
Est. Capacity (USD)	\$0
Current Market Impact	0.32 bps
Participation Rate	0.000010%